Lesson 2.5 - Line Plots

> **Video:** A Beginner's Guide To Line Plots by The Organic Chemistry Tutor **Chapter:** 2 - Graphs and Tables

Idea

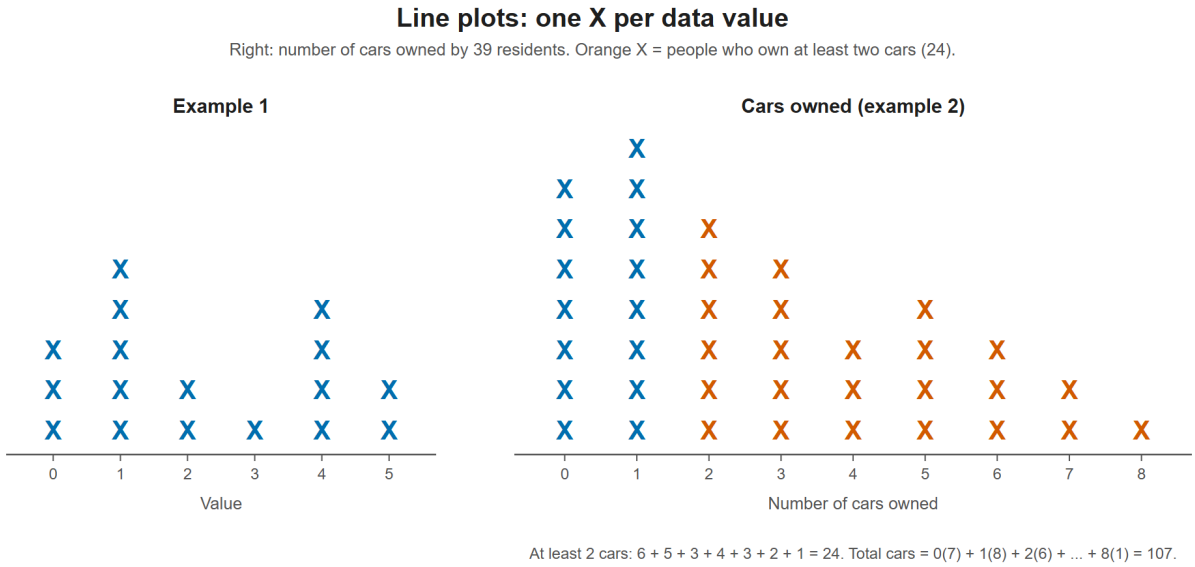
A **line plot** is the same idea as a dot plot. Draw a number line and put an **X** above a value each time it appears in the data.

Example 1 - drawing a line plot

The data contains: three 0s, five 1s, two 2s, one 3, four 4s, two 5s.

1. Sort the values: 0 0 0 1 1 1 1 1 2 2 3 4 4 4 4 5 5
2. Draw a number line from 0 to 5.
3. Stack one X per value.

```
      X
      X       X
   X  X       X
   X  X  X    X  X
   X  X  X  X X  X
----+--+--+--+--+--+--
   0  1  2  3  4  5
```



Left: the line plot from Example 1. Right: the car survey from Example 2; the orange X's are the 24 people who own at least two cars.

Example 2 - reading a line plot

A survey asked everyone in a 55-and-older community how many cars they own. The line plot shows:

Cars owned (x)	0	1	2	3	4	5	6	7	8
Number of people (X's)	7	8	6	5	3	4	3	2	1

(a) How many people own exactly one car? Count the X's above 1 → **8**.

(b) How many own *at least* two cars ($x \geq 2$)? Add everything from 2 upward:

$$\#(x \geq 2) = f_2 + f_3 + \cdots + f_8 = 6 + 5 + 3 + 4 + 3 + 2 + 1 = 24$$

(c) How many own *at most* three cars ($x \leq 3$)? Add 0 through 3, including 3:

$$\#(x \leq 3) = f_0 + f_1 + f_2 + f_3 = 7 + 8 + 6 + 5 = 26$$

(d) How many own four or five cars? $f_4 + f_5 = 3 + 4 = 7$

(e) How many people live in the community? Assuming everyone answered, add all the X's:

$$n = \sum f = 7 + 8 + 6 + 5 + 3 + 4 + 3 + 2 + 1 = 39$$

Here f_k is the number of X's above k cars.

(f) How many cars are there in total? Multiply each number of cars by how many people own that many, then add:

x	People	Cars = $x\times$ people
0	7	0
1	8	8
2	6	12
3	5	15
4	3	12
5	4	20
6	3	18
7	2	14
8	1	8
Total	**39**	**107**

There are **107 cars** in the community.

Key takeaways

- One X per data value. The height of a column is the frequency.
- **"At least k"** means $x \geq k$. **"At most k"** means $x \leq k$. Both **include** k.
- Total number of people = total number of X's.
- Total amount (cars) = $\sum(\text{value} \times \text{frequency})$, the same idea as computing a mean from a frequency table.